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N + - Network Fundamentals & Building Networks.

VLAN Trunking and Access Ports.

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TASK 1.

❖ In Cisco Packet Tracer, configure two switches (Switch1 and Switch2) and connect them with a crossover cable. Set the FastEthernet0/1 port on both switches as trunk ports using the 802.1Q protocol.

- **Configure Trunk Ports Between Two Switches:**

  Connect Switches: Switch1 and Switch2 were connected using a crossover cable through FastEthernet0/1.

  Configure Switch1:

  Switch1(config)# interface fa0/1
  Switch1(config-if)# switchport trunk encapsulation dot1q
  Switch1(config-if)# switchport mode trunk

  Configure Switch2:

  Switch2(config)# interface fa0/1
  Switch2(config-if)# switchport trunk encapsulation dot1q
  Switch2(config-if)# switchport mode trunk

  Result: Both FastEthernet0/1 ports were configured as trunk ports using IEEE 802.1Q.

TASK 2.

❖ Create three VLANs (10, 20, 30) on both switches and assign FastEthernet0/2, 0/3, and 0/4 on Switch1 to VLAN 10, 20, and 30 respectively as access ports.

- **Creating VLANs on Two Switches:**

Create VLANs on Both Switches
Run these commands on Switch1 and Switch2:

```
Switch(config)# vlan 10
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# exit
Switch(config)# vlan 30
Switch(config-vlan)# exit
```

Assign Access Ports on Switch1:

```
Switch1(config)# interface fa0/2
Switch1(config-if)# switchport mode access
Switch1(config-if)# switchport access vlan 10
Switch1(config-if)# exit
Switch1(config)# interface fa0/3
Switch1(config-if)# switchport mode access
Switch1(config-if)# switchport access vlan 20
Switch1(config-if)# exit
Switch1(config)# interface fa0/4
Switch1(config-if)# switchport mode access
Switch1(config-if)# switchport access vlan 30
```

Result: VLANs 10, 20, and 30 were created on both switches, and Switch1's ports were assigned to their respective VLANs.

TASK 3.

❖ Connect three PCs to Switch1 on ports assigned to VLAN 10, 20, and 30. Verify that PCs in different VLANs cannot communicate with each other, but can communicate with other PCs in the same VLAN.

- **Testing Communication Between VLANs:**

  Network Setup: Three PCs were connected to Switch1 on the ports assigned to VLAN 10, VLAN 20, and VLAN 30.

  Connectivity Test: Ping tests were performed between the PCs.

  - **Different VLANs:** PCs in different VLANs could not communicate because no inter VLAN routing was configured.

  - **Same VLAN:** PCs in the same VLAN can communicate if they are connected to the network and have compatible IP addresses.

  Result: The ping tests demonstrated that VLANs separate network traffic and prevent direct communication between different VLANs without a router or Layer 3 switch.

TASK 4.

❖ Change the trunk encapsulation on the trunk link between Switch1 and Switch2 from 802.1Q to ISL, and test if VLAN traffic still passes between the switches, Use the 'switchport trunk encapsulation isl' command before setting the port to trunk mode.

- **Changing Trunk Encapsulation to ISL:**

Configure Switch1:

Switch1(config)# interface fa0/1
Switch1(config-if)# switchport trunk encapsulation isl
Switch1(config-if)# switchport mode trunk

Configure Switch2:

Switch2(config)# interface fa0/1
Switch2(config-if)# switchport trunk encapsulation isl
Switch2(config-if)# switchport mode trunk

Test VLAN Traffic: Ping tests were performed between PCs in the same VLAN across both switches.

Result: VLAN traffic can pass if both switches support and use ISL on the trunk link. If the switch model does not support ISL, the command will be rejected and 802.1Q must be used instead.

TASK 5.

❖ Simulate a scenario where a new port on Switch2 (FastEthernet0/5) is accidentally set as a trunk port instead of an access port for VLAN 20. Observe and explain the impact on connectivity for a PC connected to that port.

- **Impact of Incorrect Trunk Port Configuration:**

Problem: Switch2's FastEthernet0/5 was mistakenly configured as a trunk port instead of an access port for VLAN 20.

Impact on Connectivity: The PC may experience connectivity issues because it sends regular, untagged traffic, while the trunk port is configured to carry traffic for multiple VLANs. Its traffic may be placed in the wrong VLAN.

Solution: Configure the port as an access port and assign it to VLAN 20.

Switch2(config)# interface fa0/5
Switch2(config-if)# switchport mode access
Switch2(config-if)# switchport access vlan 20

Result: The PC can communicate correctly with devices in VLAN 20, provided its IP configuration is correct.